Assignment 13

1.Scan for available Wi-Fi networks on your laptop or phone and list the SSIDs you find, noting which ones are open (unsecured) and which require a password.

Ans-

Step 1: Turn ON Wi-Fi.

- Click the Wi-Fi icon in the taskbar.
- Make sure Wi-Fi is ON.

Step 2: Open the available networks list.

- Click the Wi-Fi icon.
- You will see nearby Wi-Fi networks.

SSID / Wi-Fi Name	Security
MyHomeWiFi	Password required
JioFiber_1234	Password required
ABC_Guest	Open / Unsecured
Cafe_WiFi	Password required

Command: netsh wlan show networks

**2.Set up a mobile hotspot on your phone, connect a laptop to it, and check what IP address your laptop receives from the hotspot.

Hint: On Windows, use 'ipconfig' in Command Prompt; on Mac/Linux, use 'ifconfig' in Terminal.**

Ans-

Step 1: Open:

Settings → Connections → Mobile Hotspot and Tethering

Step 2: Tap Mobile Hotspot.

Step 3: Turn it ON.

You should see:

- Hotspot name/SSID
- Password
- Connected devices

Step 4: On the laptop, click the Wi-Fi icon.

Step 5: Select your phone's hotspot.

Example:

Amrin_Hotspot

Step 6: Enter the hotspot password.

Step 7: Confirm that the laptop is connected.

Step 8: Open **Command Prompt**.

Press: Windows + R

Type: cmd

Press **Enter**.

Step 9: Run: ipconfig

3.Research and explain in your own words what WPA2 and WPA3 are, and list two differences between them relevant to security.

Ans-

Step 1 — Understand WPA2

WPA2 = Wi-Fi Protected Access 2.

It is a Wi-Fi security standard used to protect wireless networks. WPA2 commonly uses AES encryption to protect wireless communication.

Step 2 — Understand WPA3

WPA3 = Wi-Fi Protected Access 3.

It is the newer Wi-Fi security standard designed to provide stronger protection than WPA2.

Feature	**WPA2**	**WPA3**
Authentication	Commonly uses PSK	Uses **SAE** for personal networks
Password attack protection	Weaker against offline password guessing	Better protection against offline password guessing
Security	Strong when configured correctly	Generally stronger
Newer standard	Older	Newer

4.Find a news article or real-world case from the past 3 years where a public Wi-Fi network was used in a cyber attack, and summarize how the attack happened and what could have prevented it.

Ans-

An Evil Twin attack is a malicious cyber deception where a hacker sets up a rogue Wi-Fi access point that mimics a legitimate, trusted network to intercept user traffic and harvest personal credentials.

How an Evil Twin Attack Works

- The Setup: The attacker creates a fraudulent wireless hotspot using the exact same Service Set Identifier (SSID) as a legitimate public network (such as at a cafe, hotel, or airport).
- The Trap: Devices looking for familiar or stronger signals automatically connect to the malicious access point without the user realizing anything is wrong.
- The Interception: Attackers often use a fake captive portal—a replica of a standard public Wi-Fi login page—to trick victims into entering their email, social media, or financial credentials. This functions as a sophisticated man-in-the-middle attack where all internet traffic routes directly through the hacker's hardware.

Notable 2024 Incidents

- The Australian Airport & Flight Case: In April and June 2024, the Australian Federal Police (AFP Media Release) arrested a 42-year-old West Australian man after an airline reported a suspicious in-flight Wi-Fi network.
- The Operation: The perpetrator used a portable wireless setup to deploy fake free Wi-Fi networks labeled identically to official public networks across major airports (Perth Airport, Melbourne Airport, and Adelaide Airport) as well as on active domestic commercial flights.
- The Arrest: He was charged with nine cybercrime offenses related to unauthorized data access and harvesting personal information via fraudulent landing pages. (He was later sentenced in late 2025 to over seven years in prison)